FINAL EXAM

CS372 FORMAL LANGUAGES & THE THEORY OF COMPUTATION

Student ID: _____

Name: _____

Instructions:
You have 90 minutes to complete the exam. Answer Parts A and B.
The exam will be closed book. You are not allowed to use phones and laptops too. However, you are allowed to bring a hard copy of my slides with review notes (without multiple choice questions).

PART 1: YES/NO AND MULTIPLE CHOICE QUESTIONS

YES/NO QUESTIONS

1. The set of compliments of regular languages is co-NP.
2. Language
$$L = \{a^m b^n c^{\max(m,n)} \mid m, n \in \mathbb{N}\}$$
is regular.
3. It is possible that the intersection of a finite number of regular languages is not regular.
4. Language
$$L = \{(ab)^n a^n b^n \mid n > 0\}$$
is context free.
5. Grammar
$$\{S \to 1S1;\ S \to T;\ T \to 1X1;\ T \to X;\ X \to 0X0;\ X \to 1\}$$
is ambiguous.
6. For every CFL, G, there exists a DPDA M such that $L(G) = L(M)$ and vice versa.
7. The operations of PDA never work on elements, other than the top of stack.
8. A Turing machine can simulate any algorithmic computation.
9. $$E_{\text{TM}} = \{(M)\ M \text{ is a TM and } L(M) = \varnothing\}$$
is in NP complete.
10. Ordinary Turing machines (deterministic, 1 tape) and Multitape Turing machines are equivalent in terms of the language they can recognize.

MULTIPLE CHOICE QUESTIONS

11. Which of the following components is not in the formal definition of a Deterministic Pushdown Automaton?
    A. Set of states
    B. Tape alphabet

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C. Stack alphabet

D. Input alphabet

12. Which of the following is true about regular languages?

    A. Language
    $$L = \{w \in \{0,1\}^* \mid w \text{ has an equal number of 01's and 10's}\}$$
    is regular.

    B. All infinite languages have infinite complements.

    C. Give a DFA D, there is a procedure to find a regular expression to denote $L(D)$.

    D. If L^* is regular, then L is regular.

13. Turing machine (TM) is more powerful than FMS (Finite State Machine) because

    A. tape movement is confined to one direction.

    B. it has no finite state

    C. it has the capability to remember arbitrarily long sequences of input symbols

    D. none of these

14. TM states are called the halt states:

    A. ACCEPT AND WRITE

    B. ACCEPT and READ

    C. ACCEPT AND START

    D. ACCEPT and REJECT

15. Which of the following is false about the power of

    A. DTM and NTM are the same

    B. NTM and RE are different

    C. DPDA and DTM are different

    D. DPDA and NPDA are the same

16. Let N is an NFA with n states. If we convert N to an equivalent DFA using subset construction. Here we are seeking an approximate bound on the number of states in D. Choose the best answer of the possibilities below

    A. $O(n^2)$

    B. $O(n \log n)$

    C. $O(2^n)$

    D. Larger than any of the above.

17. Consider the language
    $$L = \{w\#w^R \mid w \in \{a,b\}^*\}.$$
    Suppose we are in the process of attempting to demonstrate that L is not a regular language using the Pumping Lemma for regular languages. Which of the following strings is a suitable choice for the string z in the statement of the lemma (p is the pumping length)?

A. $aaaa\#aaaa$

B. $aaaa\#bbbb$

C. $a^p\#b^p$

D. $a^p\#a^p$

18. The language which is generated by the grammar

$$S \to aSa \mid bSb \mid \varepsilon$$

over the alphabet $\{a, b\}$ is the set of

A. Strings that begin and end with the same symbol

B. All odd and even length palindromes (sequence that reads the same backward as forward)

C. All odd length palindromes

D. All even length palindromes

19. Let

$$A = \left\{ \begin{bmatrix} 001 \\ 01 \end{bmatrix}, \begin{bmatrix} 0011 \\ 111 \end{bmatrix}, \begin{bmatrix} 11 \\ 111 \end{bmatrix}, \begin{bmatrix} 101 \\ 010 \end{bmatrix} \right\}$$

and

$$B = \left\{ \begin{bmatrix} 00 \\ 0 \end{bmatrix}, \begin{bmatrix} 001 \\ 11 \end{bmatrix}, \begin{bmatrix} 1000 \\ 0111 \end{bmatrix} \right\}.$$

are two instances of post-correspondence problem. Which of the following statements is true

A. Only A has a match

B. Both A and B have a match

C. Only B has a match

D. Neither A nor B has a match

20. To which of the following class does the NPCP problem belong?

A. P class

B. co-NP class

C. NP complete class

D. None of the above

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ANSWER SHEET

QUESTION	ANSWER
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PART II: WRITING EXAM

Question 1. Consider the following grammar:

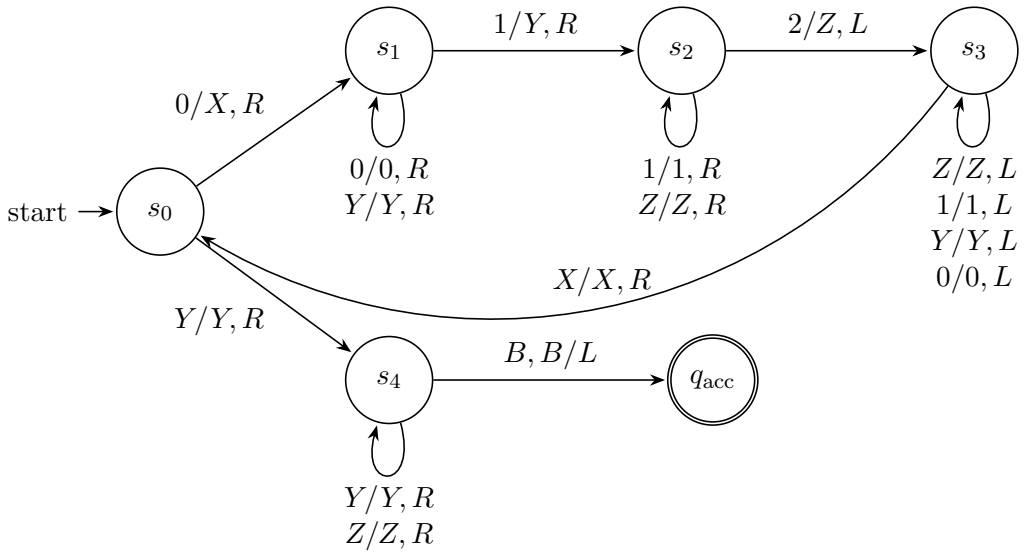
$$G: \ S \to 00S \mid S1 \mid 1$$

(a) Describe the language $L(G)$ using set notation?

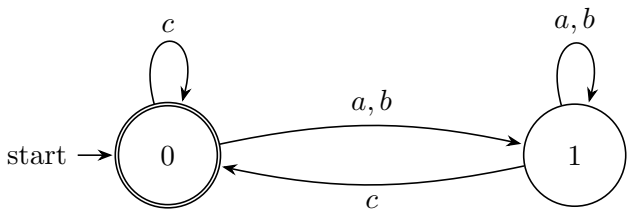
(b) Convert G to an equivalent NPDA

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Question 2: Consider the following Turing machine M. Give the sequence of configurations that M enters when started on input string 012.



Question 3: Consider the following deterministic finite automaton D



(a) Give the encoding of D

(b) Give a true instance of the problem of acceptance of a regular language with the given automaton.

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